

Eldis YMERAJ

COMPUTER VISION • DEEP LEARNING • EXPLAINABLE AI

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Profile

Junior AI profile specialized in Computer Vision, Deep Learning and Explainable AI. Currently an intern at Airbus, working on explainable AI for Vision Landing Systems and runway analysis. Looking for a full-time position in Computer Vision / Deep Learning, with a strong interest in robust, interpretable and industry-oriented visual AI systems.

Skills

Programming

Python, Java, C, C++, SQL, JavaScript

Machine Learning

Scikit-learn, SVM, PCA, clustering, GMM, boosting, optimization

Deep Learning

PyTorch, TensorFlow, Keras, CNN, transfer learning, fine-tuning

Computer Vision

OpenCV, YOLO, ByteTrack, image processing, video tracking

Explainable AI

Xplique, concept-based methods, attribution methods

NLP

Transformers, CamemBERT, BERT, Hugging Face

Data & Cloud

Pandas, NumPy, MongoDB, PostgreSQL, SQLite, AWS EC2, S3 Bucket

MLOps / Tools

Git, GitHub, Docker, Linux, Jupyter, Google Colab, LaTeX

Soft Skills

Strengths

Teamwork, autonomy, curiosity, rigor, adaptability, communication

Organization

Time management, stress management, fast learning

Languages

French

C2 – Fluent

English

C1 – Professional

Interests

Activities

Scientific reading, sports, basketball, travel, video games

Education

University of Caen Normandy

Caen, France

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE

09/2024 – 09/2026

- Supervised and unsupervised learning, Deep Learning, NLP, Computer Vision, Data Mining, ETL, SQL/NoSQL databases.

University Clermont Auvergne – ISIMA

Clermont-Ferrand, France

BACHELOR'S DEGREE IN COMPUTER SCIENCE

09/2021 – 05/2024

- Databases, artificial intelligence, applied mathematics, linear programming, advanced algorithms, probability and statistics, operations research.

Professional Experience

Airbus SAS

Toulouse, France

ARTIFICIAL INTELLIGENCE / XAI INTERN

03/2026 – 09/2026

- Contributing to the RELAI-VLS project on explainable AI for Vision Landing Systems applied to runway analysis.
- Studying and adapting explainability methods to analyze Computer Vision model decisions in an aeronautical context.
- Designing experiments to extract, visualize and compare concepts learned by object detection models.
- Analyzing internal activations, reconstruction quality and concept stability across datasets and experimental parameters.
- Tools:** Python, PyTorch, YOLO, Xplique, NumPy, Pandas, Matplotlib, AWS EC2, S3, Git.

Responsibio

Caen, France

DATA SCIENCE INTERN

06/2025 – 08/2025

- Designed an automatic classification system for mouse vocalizations to support bio-surveillance and animal welfare monitoring.
- Automated an analysis previously performed manually, reducing the time required to process audio recordings.
- Built a signal processing pipeline including feature extraction, spectral representations and data preparation.
- Trained and evaluated an SVM classifier optimized with PCA to distinguish vocalization types.
- Tools:** Python, Librosa, Scikit-learn, PyTorch, NumPy, Pandas, Git.

Projects

Research Project

Automatic Annotation of Clinical Narratives

09/2025 – 03/2026

PYTHON, PYTORCH, TRANSFORMERS, CAMEMBERT, PANDAS

- Developed an NLP pipeline to automatically annotate autobiographical patient narratives at word level.
- Prepared the corpus, used BIO annotation and fine-tuned CamemBERT to distinguish episodic and semantic details.

Handball Video Tracking

12/2024

YOLO, BYTETRACK, OPENCV, PYTHON

- Developed a Computer Vision pipeline to detect and track players, goalkeepers, referees and the ball in match videos.
- Structured the video tracking pipeline to analyze movements on the court and separate player tracking from ball tracking.

Interactive Sales Data Visualization

05/2025

DOCKER, NODE.JS, GRAPHQL, MONGODB, D3.JS, JAVASCRIPT

- Developed an interactive web application to explore sales data using a multi-container architecture.
- Implemented a GraphQL API connected to MongoDB and a dynamic visualization interface with D3.js.